University of Florida, Department of Mathematics

Logic PhD Exam, January 2009

Answer six questions and at least one from each section.

Section 1

1. Sketch the proof of Gödel's Completeness Theorem for Predicate Logic and explain the use of Henkin constants (or Skolem functions).

2. Show that for any elementary chain $\{\mathcal{A}_i\}_{i\in\omega}$ of structures, $\mathcal{A}_i$ is an elementary submodel of the union $\mathcal{A}$.

3. Show that the theory of dense linear orderings without end points is complete and decidable.

Section 2

4. Suppose $2 \leq \kappa \leq \lambda$ and λ is infinite. Show that $2^\lambda = \kappa^\lambda$.

5. State and prove König's Lemma. (Hint: trees.)

6. Show that for any notion of forcing $\mathcal{P}$ and any countable set $\mathcal{D}$ of $\mathcal{P}$-dense sets, there exists a $\mathcal{D}$-generic $\mathcal{P}$-filter.

Section 3

7. Sketch a proof of Gödel's incompleteness Theorem for Arithmetic.

8. Define many-one and truth-table reducibility and show that one implies the other but the converse does not hold in general.

9. State the Friedberg–Muchnik Theorem and sketch the proof. Explain how the "priority" argument deals with "injury".